Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **distributed systems**, concurrent pipelines, and cross-platform services using **Python, C++, Java, SQL**, and cloud infrastructure, grounded in **algorithms, operating systems**, and production testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a **multi-tenant** platform on Azure SQL and Blob Storage with row-level isolation and normalized schemas, supporting 500+ users while reducing indexed-query latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant **asynchronous orchestration** with explicit state transitions and failure isolation, processing 1K+ jobs while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Designed a **concurrent batch-processing** pipeline using multithreading and secure PDF, DOCX, and ZIP ingestion, processing 1K+ files per batch through AWS RDS and REST APIs with fault-tolerant execution.
- Implemented production Python services handling 10K+ daily requests for 150+ instructors, using CI/CD-backed workflows to reduce feedback turnaround 60% while preserving stable API performance during peak load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built a Python decision-transformer simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% while rigorously testing data transformations and model behavior across repeatable autonomous-driving benchmark scenarios.
- Architected a spatial-temporal encoder combining map geometry and trajectory context through multi-head attention, improving closed-loop planning stability 15% through systematic simulation, evaluation, regression testing, failure analysis, and debugging.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Engineered cross-platform **microservices** with automated VM builds and 20+ robust unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments.
- Optimized object-oriented API integrations and AWS Glue and S3 data flows, improving response times 30%, pipeline efficiency 68%, and long-term maintainability through reusable Factory and Singleton patterns.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellow *India*

- Evaluated distributed **consensus algorithms** and designed a quantum-resistant blockchain protocol integrating five lattice-based cryptographic primitives, reducing centralized-control and simulated quantum-risk exposure by 25% across repeatable benchmark workloads.
- Validated protocol behavior through repeatable benchmark simulations, documented findings in a peer-reviewed IEEE technical paper, and formally presented the distributed-systems research, methodology, and results at IISc Bengaluru.

Skills

Languages: Python, C++, Java, C#, JavaScript, TypeScript, SQL, Bash
Systems: Distributed Systems, Operating Systems, Concurrency, Multithreading, Data Structures, Algorithms, OOP, Microservices, REST APIs
Data & Cloud: PostgreSQL, SQL Server, MySQL, Azure SQL, AWS, Azure, S3, Blob Storage, Docker, Kubernetes, Linux
Quality: Git, CI/CD, Jenkins, GitHub Actions, PyTest, Mockito, Unit Testing, Integration Testing, Agile, Debugging, Code Reviews

Projects

ShardSafe | Python, Distributed Storage, Concurrency

- Built a transactional distributed-storage simulator with deterministic placement, XOR parity, snapshots, quotas, integrity hashes, and real-time metadata aggregates, recovering complete file reads after any single shard-node failure.

CPU-scheduling Simulator Live | C++, React.js, Operating Systems

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** with interactive visualization and dynamic waiting-time and turnaround-time calculations across 1K+ repeatable simulated process combinations and edge cases.

Chat2SQL Code | Java, TypeScript, SQL, Docker

- Built a cross-platform database workspace supporting 10+ engines, combining Java and TypeScript services with Dockerized workflows for schema exploration, SQL generation, query authoring, and structured data management.